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Golden String

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Problem Description

You are given a string **A** containing english lowercase alphabets.
You need to find if **A** contains a Golden string as a substring of itself.
A string **S** of length **N** is Golden iff:

1. N is atleast 3
2. S is only composed of 2 letters. Let's call them **p** and **q**.
3. Count of **p** in the string must be greater than $N/3$ and the other letters must all be **q**.

If the string **A** has a golden substring, return **1**, else return **0**.

Problem Constraints

 $1 \leq |A| \leq 10^5$

Input Format

The first argument is the string A.

Output Format

Return a single integer as state above.



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Q 1



Q 2



Q 3



Q 4



Q 5

Q 6

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Dp Integers

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Problem Description

You are given two non negative integers **A** and **B**, you have to make both of them equal to **0**.

You can perform the following types of operations:

- Replace both A and B with the floor (greatest integer function) of the square root of their product.
The cost of this operation is **2**.
- Divide either A or B by 2.
The cost of this operation is **1**.

You can do any operation any number of times.

Output the minimum cost to make both A and B equal to 0.

Problem Constraints

 $0 \leq A, B \leq 10^9$

Input Format

First argument is the integer A.

Second argument is the integer B.



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Q 1 ✔Q 2 ✔

Q 3 ○

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Swap It Harder

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Problem Description

You are given an array of integers **A** of size **N** with elements ranging from 0 to 9.

All elements were distributed to different groups and then array was reformed from these groups with the following rules:

1. Elements don't repeat in a group.
2. The number of groups is minimal, that is, no other way of distribution exists that can reduce the number of groups even further.

After the formation of these groups, they were merged in such a way that forms the lexicographically largest array.

Find the minimum number of neighbouring swaps needed to form this final array from the initial array **A**.

Since the answer can be large, output it modulo $10^9 + 7$.

Please note that the groups need not be contiguous.

Once the groups are formed, you are free to arrange the elements of **the group** as per your wish to achieve the lexicographically largest array.

Problem Constraints

$$1 \leq N \leq 10^5$$

$$0 \leq A[i] \leq 9$$

Input Format

The first argument is the integer array **A**.



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TWO Summation

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Problem Description

You are given two integer arrays **A** and **B** of length **N**.

You have to find the value of two summation:

$$Z = \sum_{i=1}^n \sum_{j=1}^n \max(A_i + B_j, B_i + A_j)$$

Return the value of Z modulo $10^9 + 7$.

Problem Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq A[i], B[i] \leq 10^5$$

Input Format

First argument is the integer array **A**.

Second argument is the integer array **B**.

Output Format



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Q 2

Q 3

Q 4

Q 5

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Space Race 2025

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Problem Description

The year is 2025. The space race is in full motion. Elon Musk and Jeff Bezos have already launched their rockets into space. Bill Gates on the other hand, is struggling to even get started. However, his good friend Elon, offers him his secret rocket fuel formula if Gates is able to solve a particular problem.

There are **A** planets. Jeff's space company is not very good so he can only jump from the i th planet to the $(i+1)$ th planet and $(i+2)$ th planet. Let **XJ** be the array of integers representing the number of ways to get to the i th planet if Jeff starts from the 1st planet. **XJ[0]** would be 1. Since the values can be very large, consider them all modulo **1e9+7**

Elon's space company is quite advanced compared to Jeff's. So from the i th planet, Elon's rocket can jump to all planets such that:

- For every set bit in **XJ[i]**'s binary representation, Elon's rocket when at the i th planet can jump ahead by values in the range $[2^{\text{bit}}, 2^{\text{bit}+1}]$.
- Let **XJ[i]** be 5. It's binary representation would be 101. Thus, from the i th planet, Elon's rocket will be able to jump ahead by $[2^0, 2^1] \cup [2^2, 2^3]$ where $\cup$ is union. If i was, let's say, equal to 23 and **XJ[i]** was equal to 5 then from there, the rocket can jump to all values in these ranges $[24, 25] \cup [27, 31]$.

Let **XE** be the array of integers that denote the number of ways to get to the i th planet using Elon's rocket, starting from the 1st planet. Elon asks you the calculate this array. Since the values can be very large, consider them all modulo **1e9 + 7**. **XE[0] = 1**

Problem Constraints

$1 \leq A \leq 1e5$

Input Format

The first and only argument is the integer A, denoting the number of planets.

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Shipment

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Problem Description

Little Pony owns a shipment company. There is a priority value and a weight associated with each package. To ship in a single batch with priorities $P_1, P_2, \dots, P_K$, they charge-

$$\sum_{i=1}^K P_i * (K + 1 - i)$$

Now, they have a queue of N packages to be shipped. The weight of i^{th} package is $A[i][0]$ and its priority is $A[i][1]$. They can not ship the i^{th} package until $i - 1^{\text{th}}$ package is shipped. Also, they can ship atmost B total weight of package at a time.

They can use multiple batches to ship the packages.

Help Little Pony find the maximum amount they can charge.

Problem Constraints

$$1 \leq N \leq 2 \times 10^5$$

$$1 \leq A[i][0] \leq 5000$$

$$-5000 \leq A[i][1] \leq 5000$$

$$1 \leq B \leq 10^9$$

$B \geq \max(A[i][0])$, so it is always possible to ship all items, as B is greater than maximum weight package.